

11. Compute the quartile deviation for the following data: 10, 30, 5, 12, 20, 40, 25, 15, 18.

The **Quartile Deviation (QD)** is calculated using the formula:

$$QD = \frac{Q_3 - Q_1}{2}$$

Step 1: Sort the data.

The sorted data is: 5, 10, 12, 15, 18, 20, 25, 30, 40.

The number of observations is $n = 9$.

Step 2: Calculate the First Quartile (Q_1).

Using the position formula Position of $Q_k = \frac{k(n+1)}{4}$, the position of Q_1 is $\frac{1(9+1)}{4} = 2.5$ -th term.

$$Q_1 = 2\text{nd term} + 0.5 \times (3\text{rd term} - 2\text{nd term}) = 10 + 0.5 \times (12 - 10) = 11$$

(Alternatively, using the 25th percentile definition: $Q_1 = 12$. We will proceed with the textbook formula for a small, discrete set).

Step 3: Calculate the Third Quartile (Q_3).

The position of Q_3 is $\frac{3(9+1)}{4} = 7.5$ -th term.

$$Q_3 = 7\text{th term} + 0.5 \times (8\text{th term} - 7\text{th term}) = 25 + 0.5 \times (30 - 25) = 25 + 2.5 = 27.5$$

Step 4: Compute the Quartile Deviation.

$$QD = \frac{Q_3 - Q_1}{2} = \frac{27.5 - 11}{2} = \frac{16.5}{2} = 8.25$$

The quartile deviation is 8.25.

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12. The standard deviation of n observations is σ . If each observation is multiplied by a constant k , what is the new

standard deviation? Also, if a constant c is added to each observation instead, what happens to the standard deviation?

Effect of Multiplication

If each observation is multiplied by a constant k , the new standard deviation (σ_{new}) is the absolute value of k times the original standard deviation (σ).

$$\sigma_{\text{new}} = |k|\sigma$$

The standard deviation is a measure of dispersion, and multiplication by k changes the scale of dispersion by a factor of $|k|$.

Effect of Addition

If a constant c is added to each observation, the new standard deviation (σ_{new}) **remains the same** as the original standard deviation (σ).

$$\sigma_{\text{new}} = \sigma$$

Adding a constant only shifts the entire data set, but it does not change the spread or dispersion of the data around the mean.

σ

13. Calculate the median of the following data that relates to the monthly salaries of employees (in thousand rupees): 110, 115, 108, 112, 120, 116, 140, 135, 128, 132.

The median is the middle-most value of a sorted data set.

Step 1: Sort the data.

The sorted salaries (in thousand rupees) are:

108, 110, 112, 115, 116, 120, 128, 132, 135, 140

The number of observations is $n = 10$ (an even number).

Step 2: Calculate the median.

For an even number of observations, the median is the average of the two middle terms: the $\frac{n}{2}$ -th term and the $(\frac{n}{2} + 1)$ -th term.

- $\frac{10}{2} = 5$ -th term = 116
- $(\frac{10}{2} + 1) = 6$ -th term = 120

$$\text{Median} = \frac{5\text{-th term} + 6\text{-th term}}{2} = \frac{116 + 120}{2} = \frac{236}{2} = 118$$

The median salary is 118 thousand rupees.

Q

14. In a frequency distribution, the coefficient of skewness based on the quartiles is 0.6. If the sum of the upper and the lower quartile is 100 and the median is 38, determine the values of the upper and the lower quartiles.

The formula for the **Quartile Coefficient of Skewness** (Bowley's coefficient) is:

$$S_k = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

We are given:

- $S_k = 0.6$
- $Q_3 + Q_1 = 100$
- Median (Q_2) = 38

Step 1: Substitute the given values into the formula to find the difference of the quartiles.

$$0.6 = \frac{100 - 2(38)}{Q_3 - Q_1}$$

$$0.6 = \frac{100 - 76}{Q_3 - Q_1}$$

$$0.6 = \frac{24}{Q_3 - Q_1}$$

Rearrange to solve for $(Q_3 - Q_1)$:

$$Q_3 - Q_1 = \frac{24}{0.6} = 40$$

Step 2: Solve the system of two linear equations.

We have:

$$1. Q_3 + Q_1 = 100$$

$$2. Q_3 - Q_1 = 40$$

Add Equation (1) and Equation (2):

$$(Q_3 + Q_1) + (Q_3 - Q_1) = 100 + 40$$

$$2Q_3 = 140$$

$$Q_3 = 70$$

Substitute $Q_3 = 70$ back into Equation (1):

$$70 + Q_1 = 100$$

$$Q_1 = 100 - 70$$

$$Q_1 = 30$$

The upper quartile (Q_3) is 70 and the lower quartile (Q_1) is 30.

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15. The standard deviation of a symmetric distribution is 4. Calculate the value of μ_4 so that the distribution must be mesokurtic.

A distribution is defined as **mesokurtic** if its coefficient of kurtosis (β_2) is equal to 3.

The relationship between the fourth central moment (μ_4) and the second central moment (μ_2) is:

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

We are given:

- Standard deviation (σ) = 4.
- For a mesokurtic distribution, $\beta_2 = 3$.

Step 1: Calculate the second central moment (μ_2).

The second central moment is the variance (σ^2).

$$\mu_2 = \sigma^2 = 4^2 = 16$$

Step 2: Calculate μ_4 for $\beta_2 = 3$.

$$3 = \frac{\mu_4}{\mu_2^2}$$

$$3 = \frac{\mu_4}{16^2}$$

$$3 = \frac{\mu_4}{256}$$

$$\mu_4 = 3 \times 256 = 768$$

The value of μ_4 must be 768 for the distribution to be mesokurtic.

Q

16. A bus runs at a speed of 60 kph over 50 kilometers; the next 30 kilometers at a speed of 40 kph; the next 20 kilometers at a speed of 30 kph; and the final run of 50 kilometers at a speed of 25 kph. Find the average speed in kilometers per hour.

The **average speed** is calculated by the formula:

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

Step 1: Calculate the Total Distance.

$$\text{Total Distance} = 50 + 30 + 20 + 50 = \mathbf{150 \text{ km}}$$

Step 2: Calculate the Total Time.

Time is calculated as $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$.

- $T_1 = \frac{50 \text{ km}}{60 \text{ kph}} = \frac{5}{6} \text{ hours}$
- $T_2 = \frac{30 \text{ km}}{40 \text{ kph}} = \frac{3}{4} \text{ hours}$
- $T_3 = \frac{20 \text{ km}}{30 \text{ kph}} = \frac{2}{3} \text{ hours}$
- $T_4 = \frac{50 \text{ km}}{25 \text{ kph}} = 2 \text{ hours}$

$$\text{Total Time} = T_1 + T_2 + T_3 + T_4 = \frac{5}{6} + \frac{3}{4} + \frac{2}{3} + 2$$

To sum the fractions, use the common denominator 12:

$$\text{Total Time} = \frac{10}{12} + \frac{9}{12} + \frac{8}{12} + \frac{24}{12} = \frac{10 + 9 + 8 + 24}{12} = \frac{51}{12} \text{ hours}$$

Step 3: Compute the Average Speed.

$$\text{Average Speed} = \frac{150 \text{ km}}{\frac{51}{12} \text{ hours}} = 150 \times \frac{12}{51}$$

$$\text{Average Speed} = \frac{1800}{51}$$

Simplify the fraction by dividing the numerator and denominator by their greatest common divisor, 3:

$$\text{Average Speed} = \frac{1800 \div 3}{51 \div 3} = \frac{600}{17} \text{ kph}$$

As a decimal, this is approximately:

$$\text{Average Speed} \approx 35.29 \text{ kph}$$

The average speed is $\frac{600}{17}$ kph (or approximately 35.29 kph).

17. Moments About the Mean, Skewness, and Kurtosis

This problem requires converting the moments about an arbitrary point (working mean) to moments about the actual mean, and then calculating the measures of skewness (β_1) and kurtosis (β_2).

Given Data

- Working Mean (A) = 28.5
- Moments about A (μ'_k):
 - $\mu'_1 = c = 0.294$
 - $\mu'_2 = 7.144$
 - $\mu'_3 = 42.409$
 - $\mu'_4 = 454.98$

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Part 1: Moments About the Mean (μ_k)

We use the conversion formulas, where $c = \mu'_1 = 0.294$.

1. First Moment (μ_1)

The first moment about the mean is always zero.

$$\mu_1 = 0$$

2. Second Moment (μ_2)

$$\mu_2 = \mu'_2 - c^2$$

$$\mu_2 = 7.144 - (0.294)^2$$

$$\mu_2 = 7.144 - 0.086436 = \mathbf{7.057564}$$

(μ_2 is the **Variance**).

3. Third Moment (μ_3)

$$\mu_3 = \mu'_3 - 3\mu'_2c + 2c^3$$

$$\mu_3 = 42.409 - 3(7.144)(0.294) + 2(0.294)^3$$

$$\mu_3 = 42.409 - 6.295488 + 0.050800 \approx \mathbf{36.164312}$$

(Using the exact intermediate result from the calculation: $\mu_3 \approx 36.158816$)

4. Fourth Moment (μ_4)

$$\mu_4 = \mu'_4 - 4\mu'_3c + 6\mu'_2c^2 - 3c^4$$

$$\mu_4 = 454.98 - 4(42.409)(0.294) + 6(7.144)(0.294)^2 - 3(0.294)^4$$

$$\mu_4 = 454.98 - 49.882488 + 3.682390 - 0.023533 \approx \mathbf{408.756369}$$

(Using the exact intermediate result from the calculation: $\mu_4 \approx 408.789595$)

Moment about the Mean	Value
μ_1	0
μ_2	7.0576
μ_3	36.1588
μ_4	408.7896

or

Part 2: Measures of Skewness

The measure of skewness is given by β_1 (or its square root, γ_1).

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

$$\beta_1 = \frac{(36.158816)^2}{(7.057564)^3} = \frac{1307.458}{351.464} \approx \mathbf{3.7193}$$

Interpretation: Since $\beta_1 > 0$, the distribution is **positively skewed**.

σ

Part 3: Measures of Kurtosis

The measure of kurtosis is given by β_2 .

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

$$\beta_2 = \frac{408.789595}{(7.057564)^2} = \frac{408.789595}{49.809226} \approx \mathbf{8.2071}$$

Interpretation: Since $\beta_2 > 3$, the distribution is **leptokurtic** (more peaked and heavier-tailed than a normal distribution).

Measure	Value
Skewness (β_1)	3.7193
Kurtosis (β_2)	8.2071

18. Find the First Four Moments About the Origin

This problem requires calculating the first four moments about the origin (μ'_k) using the given mean, variance, and measures of skewness (γ_1) and kurtosis (β_2).

Given Data

- Mean ($\bar{X}$) = 10

- Variance (μ_2) = **16**
- Skewness (γ_1) = **1**
- Kurtosis (β_2) = **4**

Step 1: Calculate μ'_1

The first moment about the origin is equal to the mean.

$$\mu'_1 = \bar{X} = \mathbf{10}$$

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Step 2: Calculate μ_3 and μ_4 (Moments about the Mean)

We need the third (μ_3) and fourth (μ_4) moments about the mean before converting them to moments about the origin.

A. Third Moment (μ_3)

The skewness coefficient γ_1 is related to μ_3 and μ_2 by:

$$\gamma_1 = \frac{\mu_3}{\sqrt{\mu_2^3}} \implies \mu_3 = \gamma_1 \cdot (\mu_2)^{3/2}$$

$$\mu_3 = 1 \cdot (16)^{3/2} = 1 \cdot (\sqrt{16})^3 = 1 \cdot 4^3 = \mathbf{64}$$

B. Fourth Moment (μ_4)

The kurtosis coefficient β_2 is related to μ_4 and μ_2 by:

$$\beta_2 = \frac{\mu_4}{\mu_2^2} \implies \mu_4 = \beta_2 \cdot \mu_2^2$$

$$\mu_4 = 4 \cdot (16)^2 = 4 \cdot 256 = \mathbf{1024}$$

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Step 3: Calculate μ'_2, μ'_3, μ'_4 (Moments about the Origin)

We use the conversion formulas where $c = \mu'_1 = 10$.

A. Second Moment (μ'_2)

$$\begin{aligned}\mu'_2 &= \mu_2 + (\mu'_1)^2 \\ \mu'_2 &= 16 + (10)^2 = 16 + 100 = \mathbf{116}\end{aligned}$$

B. Third Moment (μ'_3)

$$\begin{aligned}\mu'_3 &= \mu_3 + 3\mu_2\mu'_1 + (\mu'_1)^3 \\ \mu'_3 &= 64 + 3(16)(10) + (10)^3 \\ \mu'_3 &= 64 + 480 + 1000 = \mathbf{1544}\end{aligned}$$

C. Fourth Moment (μ'_4)

$$\begin{aligned}\mu'_4 &= \mu_4 + 4\mu_3\mu'_1 + 6\mu_2(\mu'_1)^2 + (\mu'_1)^4 \\ \mu'_4 &= 1024 + 4(64)(10) + 6(16)(10)^2 + (10)^4 \\ \mu'_4 &= 1024 + 2560 + 9600 + 10000 = \mathbf{23184}\end{aligned}$$

Summary of Moments About the Origin

The first four moments about the origin are:

Moment	Value
μ'_1	10
μ'_2	116
μ'_3	1544
μ'_4	23184

Question 19: Calculate the Mode using Grouping

Size of Items (X)	8	9	10	11	12	13	14	15
No. of Items (f)	5	6	8	7	9	8	9	6

or

Step 1: Prepare the Grouping Table

We create five additional columns to group the frequencies:

- **Col 2:** Grouping frequencies in twos (starting from the first frequency).
- **Col 3:** Grouping frequencies in twos (starting from the second frequency).
- **Col 4:** Grouping frequencies in threes (starting from the first frequency).
- **Col 5:** Grouping frequencies in threes (starting from the second frequency).
- **Col 6:** Grouping frequencies in threes (starting from the third frequency).

X	f (Col 1)	Col 2	Col 3	Col 4	Col 5	Col 6
8	5	11 (5+6)		19 (5+6+8)		
9	6		14 (6+8)		21 (6+8+7)	
10	8	15 (8+7)				24 (8+7+9)
11	7		16 (7+9)	24 (7+9+8)		
12	9	17 (9+8)			26 (9+8+9)	

X	f (Col 1)	Col 2	Col 3	Col 4	Col 5	Col 6
13	8		17 (8+9)			23 (8+9+6)
14	9	15 (9+6)		23 (9+6)		
15	6					

The largest sum in each column is **bolded**.



Step 2: Prepare the Analysis Table

We record which items (X values) contributed to the largest sum in each column.

Column	Largest Sum	Items Covered
1 (f)	9	12, 14
2 (2s from 1st)	17	12, 13
3 (2s from 2nd)	17	13, 14
4 (3s from 1st)	24	11, 12, 13
5 (3s from 2nd)	26	12, 13, 14
6 (3s from 3rd)	24	10, 11, 12



Step 3: Count the Highest Frequency

We count how many times each item appears as a component of the largest group sums.

Item (X)	8	9	10	11	12	13	14	15	Total Tally
Col 1					1		1		2
Col 2					1	1			2
Col 3						1	1		2
Col 4				1	1	1			3
Col 5					1	1	1		3
Col 6			1	1	1				3
Total Tally	0	0	1	2	5	4	3	0	

The item with the highest total tally is the mode.

- The item **12** has the highest tally of **5**.

Conclusion

The mode calculated using the grouping method is **12**.

Note on Bimodality: The initial inspection suggested bimodality (12 and 14). However, the grouping method consolidates the frequencies and shows that the item **12** is involved in more high-frequency groupings (5 tallies) than the item **14** (3 tallies). Therefore, the actual mode is **12**.

20. Mode and Standard Deviation for Grouped Data

This problem asks for the **Mode** and the **Standard Deviation** (σ) for the given inclusive grouped data.

Marks	No. of Candidates (f)
1-10	3
11-20	16
21-30	26
31-40	31

Marks	No. of Candidates (f)
41-50	16
51-60	8
Total	$N = 100$

8

Part 1: Calculation of the Mode (Z)

Since the data is inclusive, we first convert the limits to **exclusive** limits by applying a continuity correction of 0.5.

1. Identify the Modal Class and Parameters

- The **highest frequency** is **31**, corresponding to the class **31 – 40**.
- L (Lower Boundary of exclusive class) = $31 - 0.5 = \mathbf{30.5}$
- h (Class Width) = $40.5 - 30.5 = \mathbf{10}$
- f_1 (Modal Frequency) = **31**
- f_0 (Preceding Frequency) = **26**
- f_2 (Succeeding Frequency) = **16**

2. Apply the Mode Formula

$$\text{Mode } (Z) = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

$$Z = 30.5 + \frac{31 - 26}{2(31) - 26 - 16} \times 10$$

$$Z = 30.5 + \frac{5}{62 - 42} \times 10$$

$$Z = 30.5 + \frac{5}{20} \times 10$$

$$Z = 30.5 + \frac{1}{4} \times 10 = 30.5 + 2.5 = \mathbf{33.0}$$

The Mode is 33.0 marks.

σ

Part 2: Calculation of Standard Deviation (σ)

We use the **step-deviation method** for efficiency. Let the assumed mean A be the midpoint of the modal class: $A = 35.5$.

Marks	f	Midpoint (x)	$d' = \frac{x-35.5}{10}$	fd'	$f(d')^2$
1-10	3	5.5	-3	-9	27
11-20	16	15.5	-2	-32	64
21-30	26	25.5	-1	-26	26
31-40	31	35.5 (A)	0	0	0
41-50	16	45.5	1	16	16
51-60	8	55.5	2	16	32
Total	= 100			$\sum fd' = -35$	$\sum f(d')^2 = 165$

1. Apply the Standard Deviation Formula

$$\sigma = \sqrt{\frac{\sum f(d')^2}{N} - \left(\frac{\sum fd'}{N}\right)^2} \times h$$

$$\sigma = \sqrt{\frac{165}{100} - \left(\frac{-35}{100}\right)^2} \times 10$$

$$\sigma = \sqrt{1.65 - 0.1225} \times 10$$

$$\sigma = \sqrt{1.5275} \times 10$$

$$\sigma \approx 1.2359 \times 10 = \mathbf{12.36}$$

The Standard Deviation is **12.36 marks** (rounded to two decimal places).

21. Check the Skewness of the Following Two Series

Skewness measures the degree to which a distribution is asymmetrical. We'll use **Pearson's First Coefficient of Skewness (S_k)** to check the skewness:

$$S_k = \frac{\text{Mean} - \text{Mode}}{\text{S.D.}} \quad \text{R (often used as an approximation)}$$

$$S_k \approx \frac{3(\text{Mean} - \text{Median})}{\text{S.D.}}$$

We will use the second formula since the median is given.

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Series (a)

Measure	Value
Mean	32
Median	34
S.D.	20

Calculate Skewness:

$$S_k = \frac{3(\text{Mean} - \text{Median})}{\text{S.D.}} = \frac{3(32 - 34)}{20}$$

$$S_k = \frac{3(-2)}{20} = \frac{-6}{20} = -\mathbf{0.3}$$

Conclusion for Series (a):

Since the coefficient of skewness (S_k) is **negative** (-0.3), the distribution of Series (a) is **Negatively Skewed** (or skewed to the left).

Series (b)

Measure	Value
Mean	32
Median	36
S.D.	25

Calculate Skewness:

$$S_k = \frac{3(\text{Mean} - \text{Median})}{\text{S.D.}} = \frac{3(32 - 36)}{25}$$

$$S_k = \frac{3(-4)}{25} = \frac{-12}{25} = -0.48$$

Conclusion for Series (b):

Since the coefficient of skewness (S_k) is **negative** (-0.48), the distribution of Series (b) is also **Negatively Skewed**.

Summary

- **Series (a):** Negatively Skewed ($S_k = -0.3$)
- **Series (b):** Negatively Skewed ($S_k = -0.48$)

22. First Four Moments About the Mean and Skewness

This problem requires calculating the first four moments about the mean (μ_k) and the measure of skewness (β_1) for the given discrete frequency distribution.

X	2	3	4	5	6
f	1	3	7	3	1

Step 1: Calculate the Mean ($\bar{X}$)

To find the moments about the mean, we first need to calculate the mean.

$$\bar{X} = \frac{\sum fX}{\sum f}$$

X	f	fX
2	1	2
3	3	9
4	7	28
5	3	15
6	1	6
Total	$\sum f = 15$	$\sum fX = 60$

$$\bar{X} = \frac{60}{15} = 4$$

Step 2: Calculate Moments About the Mean (μ_k)

The central moment formula is: $\mu_k = \frac{\sum f(X-\bar{X})^k}{\sum f}$.

Here, the deviation $d = X - 4$.

X	f	$d = X - 4$	fd	fd^2	fd^3	fd^4
2	1	-2	-2	4	-8	16
3	3	-1	-3	3	-3	3

X	f	$d = X - 4$	fd	fd^2	fd^3	fd^4
4	7	0	0	0	0	0
5	3	1	3	3	3	3
6	1	2	2	4	8	16
Total	= 15		0	14	0	38

1. First Moment (μ_1):

$$\mu_1 = \frac{\sum fd}{N} = \frac{0}{15} = 0$$

(This is always true for the first moment about the mean).

2. Second Moment (μ_2):

$$\mu_2 = \frac{\sum fd^2}{N} = \frac{14}{15} \approx \mathbf{0.9333}$$

3. Third Moment (μ_3):

$$\mu_3 = \frac{\sum fd^3}{N} = \frac{0}{15} = 0$$

4. Fourth Moment (μ_4):

$$\mu_4 = \frac{\sum fd^4}{N} = \frac{38}{15} \approx \mathbf{2.5333}$$

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Step 3: Calculate β_1 and Check for Symmetry

The measure of skewness is β_1 :

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

$$\beta_1 = \frac{(0)^2}{(14/15)^3} = 0$$

Conclusion on Skewness:

Since $\mu_3 = 0$ and consequently $\beta_1 = 0$, the distribution is **symmetrical**. This is also confirmed by the frequencies being symmetrical around the mean (the frequencies 1, 3, 7, 3, 1 are symmetric about $X = 4$).

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Summary of Results

Moment	Value
μ_1	0
μ_2	$14/15 \approx 0.9333$
μ_3	0
μ_4	$38/15 \approx 2.5333$
β_1	0

Is it a symmetrical distribution? Yes, because $\mu_3 = 0$ and $\beta_1 = 0$.

I see you're ready for question **23** now!

23. Moments About the Mean and the Origin

This problem involves converting moments from an arbitrary value ($A = 2$) to the **mean** (μ_k) and then to the **origin** (μ'_k).

Given Data

- Arbitrary value (A) = **2**
- Moments about A (μ'_k):
 - $\mu'_1(\text{about } 2) = c = \mathbf{1}$
 - $\mu'_2(\text{about } 2) = \mathbf{2.5}$

- $\mu'_3(\text{about } 2) = 5.5$
- $\mu'_4(\text{about } 2) = 16$

μ'

Part 1: Moments About the Mean (μ_k)

We use the conversion formulas, where $c = \mu'_1 = 1$.

1. First Moment (μ_1)

$$\mu_1 = 0$$

2. Second Moment (μ_2)

$$\mu_2 = \mu'_2 - c^2$$

$$\mu_2 = 2.5 - (1)^2 = 2.5 - 1 = 1.5$$

3. Third Moment (μ_3)

$$\mu_3 = \mu'_3 - 3\mu'_2c + 2c^3$$

$$\mu_3 = 5.5 - 3(2.5)(1) + 2(1)^3$$

$$\mu_3 = 5.5 - 7.5 + 2 = 0$$

4. Fourth Moment (μ_4)

$$\mu_4 = \mu'_4 - 4\mu'_3c + 6\mu'_2c^2 - 3c^4$$

$$\mu_4 = 16 - 4(5.5)(1) + 6(2.5)(1)^2 - 3(1)^4$$

$$\mu_4 = 16 - 22 + 15 - 3 = 6$$

Moments about the Mean	Value
μ_1	0
μ_2	1.5
μ_3	0
μ_4	6

or

Part 2: Moments About the Origin (μ'_k)

The origin is the value 0. The first moment about the origin is the mean ($\bar{X}$).

1. Mean ($\bar{X}$)

The mean is calculated from the first moment about the arbitrary point A :

$$\bar{X} = A + \mu'_1 = 2 + 1 = 3$$

Therefore, $\mu'_1(\text{about } 0) = 3$.

2. Second Moment (μ'_2)

We use the formula $\mu'_2 = \mu_2 + \bar{X}^2$:

$$\mu'_2 = 1.5 + (3)^2 = 1.5 + 9 = 10.5$$

3. Third Moment (μ'_3)

We use the formula $\mu'_3 = \mu_3 + 3\mu_2\bar{X} + \bar{X}^3$:

$$\mu'_3 = 0 + 3(1.5)(3) + (3)^3$$

$$\mu'_3 = 0 + 13.5 + 27 = \mathbf{40.5}$$

4. Fourth Moment (μ'_4)

We use the formula $\mu'_4 = \mu_4 + 4\mu_3\bar{X} + 6\mu_2\bar{X}^2 + \bar{X}^4$:

$$\mu'_4 = 6 + 4(0)(3) + 6(1.5)(3)^2 + (3)^4$$

$$\mu'_4 = 6 + 0 + 6(1.5)(9) + 81$$

$$\mu'_4 = 6 + 81 + 81 = \mathbf{168}$$

Moments about the Origin	Value
μ'_1	3
μ'_2	10.5
μ'_3	40.5
μ'_4	168

I'll solve Question **24** first, which requires calculating the **moment coefficient of skewness** (γ_1) and **moment coefficient of kurtosis** (γ_2).

24. Moment Coefficients of Skewness and Kurtosis

The data is a discrete frequency distribution. We will use the **step deviation method** to calculate the first four moments about the mean (μ_k).

Height (X)	f
59	0
61	2
63	6
65	20
67	40
69	20
71	8

Height (X)	f
73	2
75	2
Total	$N = 100$

Step 1: Prepare the Calculation Table and Find Moments about an Assumed Mean

Let the Assumed Mean A be the central value, $A = 67$.

The common difference (step size) h is 2.

Let $d' = \frac{X-A}{h} = \frac{X-67}{2}$.

X	f	d'	fd'	fd'^2	fd'^3	fd'^4
59	0	-4	0	0	0	0
61	2	-3	-6	18	-54	162
63	6	-2	-12	24	-48	96
65	20	-1	-20	20	-20	20
67 (A)	40	0	0	0	0	0
69	20	1	20	20	20	20
71	8	2	16	32	64	128
73	2	3	6	18	54	162
75	2	4	8	32	128	512
Total	= 100		6	164	144	1100

Moments about the Assumed Mean (A)

$$\mu'_k(\text{in } d') = \frac{\sum fd'^k}{N}$$

- $\mu'_1 = \frac{6}{100} = 0.06$
- $\mu'_2 = \frac{164}{100} = 1.64$
- $\mu'_3 = \frac{144}{100} = 1.44$

- $\mu'_4 = \frac{1100}{100} = 11.00$

}

Step 2: Convert to Moments about the Mean (μ_k)

We use the conversion formulas, where $c = \mu'_1 = 0.06$.

1. Second Moment (μ_2)

$$\mu_2 = h^2[\mu'_2 - c^2]$$

$$\mu_2 = 2^2[1.64 - (0.06)^2] = 4[1.64 - 0.0036]$$

$$\mu_2 = 4 \times 1.6364 = \mathbf{6.5456}$$

2. Third Moment (μ_3)

$$\mu_3 = h^3[\mu'_3 - 3\mu'_2c + 2c^3]$$

$$\mu_3 = 2^3[1.44 - 3(1.64)(0.06) + 2(0.06)^3]$$

$$\mu_3 = 8[1.44 - 0.2952 + 0.000432]$$

$$\mu_3 = 8 \times 1.145232 \approx \mathbf{9.161856}$$

3. Fourth Moment (μ_4)

$$\mu_4 = h^4[\mu'_4 - 4\mu'_3c + 6\mu'_2c^2 - 3c^4]$$

$$\mu_4 = 2^4[11.00 - 4(1.44)(0.06) + 6(1.64)(0.06)^2 - 3(0.06)^4]$$

$$\mu_4 = 16[11.00 - 0.3456 + 0.035424 - 0.00003888]$$

$$\mu_4 = 16 \times 10.689785 \approx \mathbf{171.03656}$$

}

Step 3: Calculate the Moment Coefficients

1. Moment Coefficient of Skewness (γ_1)

We use the measure β_1 and $\gamma_1 = \sqrt{\beta_1}$.

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

$$\beta_1 = \frac{(9.161856)^2}{(6.5456)^3} = \frac{83.9406}{280.4578} \approx 0.2993$$

$$\gamma_1 = \sqrt{\beta_1} \approx \sqrt{0.2993} \approx \mathbf{0.5471}$$

2. Moment Coefficient of Kurtosis (γ_2)

We use the measure β_2 and $\gamma_2 = \beta_2 - 3$.

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

$$\beta_2 = \frac{171.03656}{(6.5456)^2} = \frac{171.03656}{42.84539} \approx 3.9918$$

$$\gamma_2 = \beta_2 - 3 = 3.9918 - 3 \approx \mathbf{0.9918}$$

Final Answer:

Measure	Value	Conclusion
Moment Coefficient of Skewness (γ_1)	0.5471	Positively Skewed
Moment Coefficient of Kurtosis (γ_2)	0.9918	Leptokurtic ($\beta_2 > 3$)

25. Mean, Median, and Quartiles for a Cumulative Frequency Distribution

The given data is a **"Less Than" cumulative frequency (c.f.) distribution**. We must first convert it into a simple frequency distribution with class intervals.

Step 1: Convert to Simple Frequency Distribution

Marks (Exclusive Class)	No. of Students (c.f.)	Frequency (<i>f</i>)
0-10	15	15
10-20	35	$35 - 15 = 20$
20-30	60	$60 - 35 = 25$
30-40	84	$84 - 60 = 24$
40-50	94	$94 - 84 = 10$
50-60	127	$127 - 94 = 33$
60-70	198	$198 - 127 = 71$
70-80	249	$249 - 198 = 51$
Total		= 249

↻

Step 2: Calculate the Mean ($\bar{X}$)

We use the mid-point (*x*) method, taking the Assumed Mean $A = 35$ for calculation simplicity.

Class	<i>f</i>	Midpoint (<i>x</i>)	$d' = \frac{x-35}{10}$	fd'
0-10	15	5	-3	-45
10-20	20	15	-2	-40
20-30	25	25	-1	-25
30-40	24	35 (A)	0	0
40-50	10	45	1	10

Class	f	Midpoint (x)	$d' = \frac{x-35}{10}$	fd'
50-60	33	55	2	66
60-70	71	65	3	213
70-80	51	75	4	204
Total	= 249			$\sum fd' = 383$

$$\bar{X} = A + \left(\frac{\sum fd'}{N} \right) \times h$$

$$\bar{X} = 35 + \left(\frac{383}{249} \right) \times 10$$

$$\bar{X} = 35 + 1.53815 \times 10 = 35 + 15.3815 \approx \mathbf{50.38}$$

The Mean is 50.38 marks.

»

Step 3: Calculate Median (Q_2), Lower Quartile (Q_1), and Upper Quartile (Q_3)

We use the c.f. table, where $N = 249$. The formula for quartiles/median is:

$$Q_k = L + \frac{\frac{kN}{4} - \text{c.f. precedin}}{f_{Q_k}} \times h$$

Class	f	c.f.
0-10	15	15
10-20	20	35
20-30	25	60
30-40	24	84
40-50	10	94
50-60	33	127
60-70	71	198

Class	f	c.f.
70-80	51	249

A. Lower Quartile (Q_1)

- **Position:** $N/4 = 249/4 = 62.25$.
- **Q_1 Class:** The class where c.f. is first greater than 62.25 is **30-40**.
- **Parameters:** $L = 30$, c.f._{precedin} = 60, $f_{Q_1} = 24$, $h = 10$.

$$Q_1 = 30 + \frac{62.25 - 60}{24} \times 10$$

$$Q_1 = 30 + \frac{2.25}{24} \times 10 = 30 + 0.9375 \approx \mathbf{30.94}$$

B. Median (Q_2)

- **Position:** $2N/4 = N/2 = 249/2 = 124.5$.
- **Median Class:** The class where c.f. is first greater than 124.5 is **50-60**.
- **Parameters:** $L = 50$, c.f._{precedin} = 94, $f_{Q_2} = 33$, $h = 10$.

$$Q_2 = 50 + \frac{124.5 - 94}{33} \times 10$$

$$Q_2 = 50 + \frac{30.5}{33} \times 10 = 50 + 9.2424 \approx \mathbf{59.24}$$

C. Upper Quartile (Q_3)

- **Position:** $3N/4 = 3(249)/4 = 186.75$.
- **Q_3 Class:** The class where c.f. is first greater than 186.75 is **60-70**.
- **Parameters:** $L = 60$, c.f._{precedin} = 127, $f_{Q_3} = 71$, $h = 10$.

$$Q_3 = 60 + \frac{186.75 - 127}{71} \times 10$$

$$Q_3 = 60 + \frac{59.75}{71} \times 10 = 60 + 8.4155 \approx \mathbf{68.42}$$

»

Final Results

- **Mean ($\bar{X}$): 50.38** marks
- **Lower Quartile (Q_1): 30.94** marks
- **Median (Q_2): 59.24** marks
- **Upper Quartile (Q_3): 68.42** marks

26. Determine Missing Frequencies using the Median Formula

The goal is to find the two missing frequencies (f_3 and f_5) in the incomplete frequency distribution, given that the **median is 46** and the total frequency is $N = 229$.

Variable	10-20	20-30	30-40	40-50	50-60	60-70	70-80	Total
Frequency (f)	12	30	f_3	65	f_5	25	18	229

Step 1: Set up the Cumulative Frequency (c.f.) and the Total Frequency Equation

Let $f_3 = x$ and $f_5 = y$.

Class	Frequency (f)	Cumulative Frequency (c.f.)
10-20	12	12
20-30	30	42
30-40	x	$42 + x$

Class	Frequency (f)	Cumulative Frequency (c.f.)
40-50	65	$107 + x$
50-60	y	$107 + x + y$
60-70	25	$132 + x + y$
70-80	18	$150 + x + y$

The total frequency is given as 229.

$$\sum f = 150 + x + y = 229$$

$$x + y = 229 - 150$$

$$x + y = 79 \quad (\text{Equation 1})$$

or

Step 2: Use the Median Formula to find the value of x

Given that the **Median (M) is 46**. Since 46 falls in the class **40-50**, this is the **Median Class**.

The formula for the median is:

$$M = L + \frac{\frac{N}{2} - \text{c.f.}_{\text{precedin}}}{f_M} \times h$$

Where:

- $M = 46$
- L (Lower boundary of median class) = **40**
- $N/2 = 229/2 = \mathbf{114.5}$
- $\text{c.f.}_{\text{precedin}}$ (c.f. of preceding class) = **$42 + x$**
- f_M (Frequency of median class) = **65**
- h (Class width) = $50 - 40 = \mathbf{10}$

Substitute these values into the formula:

$$46 = 40 + \frac{114.5 - (42 + x)}{65} \times 10$$

$$46 - 40 = \frac{114.5 - 42 - x}{65} \times 10$$

$$6 = \frac{72.5 - x}{65} \times 10$$

Now, solve for x :

$$6 \times 65 = (72.5 - x) \times 10$$

$$390 = 725 - 10x$$

$$10x = 725 - 390$$

$$10x = 335$$

$$x = 33.5$$

Since frequency must be an integer, we take $x = 34$ (this is a common situation in textbook problems where the calculated frequency is rounded to the nearest integer).

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Step 3: Find the value of y

Substitute the value of $x = 34$ back into Equation 1:

$$x + y = 79$$

$$34 + y = 79$$

$$y = 79 - 34$$

$$y = 45$$

Final Answer:

The missing frequencies are:

- f_3 (Frequency of 30-40 class) ≈ 34
- f_5 (Frequency of 50-60 class) $= 45$

27. Check the Consistency of the Consumption Data

Consistency in statistical data is measured using the **Coefficient of Variation (C.V.)**. The series with the **lower C.V.** is considered **more consistent** (less variable).

The formula for C.V. is:

$$\text{C.V.} = \frac{\text{Standard Deviation}(\sigma)}{\text{Mean}(\bar{X})} \times 100$$

1. Calculations for Guavas (Series X)

- Data (X): 3, 5, 6, 4, 3, 5, 4
- $N = 7$

X	X^2
3	9
5	25
6	36
4	16
3	9
5	25
4	16
$\sum X = 30$	$\sum X^2 = 136$

Mean ($\bar{X}$):

$$\bar{X} = \frac{\sum X}{N} = \frac{30}{7} \approx 4.286$$

Standard Deviation (σ_X):

$$\sigma_X = \sqrt{\frac{\sum X^2}{N} - \bar{X}^2}$$

$$\sigma_X = \sqrt{\frac{136}{7} - (4.286)^2} \approx \sqrt{19.4286 - 18.3697} \approx \sqrt{1.0589} \approx 1.030$$

Coefficient of Variation (C.V._X):

$$\text{C.V.}_X = \frac{1.030}{4.286} \times 100 \approx \mathbf{24.04}$$

»

2. Calculations for Oranges (Series Y)

- Data (Y): 1, 3, 7, 9, 2, 6, 2
- $N = 7$

Y	Y ²
1	1
3	9
7	49
9	81
2	4
6	36
2	4
$\sum Y = \mathbf{30}$	$\sum Y^2 = \mathbf{184}$

Mean ($\bar{Y}$):

$$\bar{Y} = \frac{\sum Y}{N} = \frac{30}{7} \approx 4.286$$

Standard Deviation (σ_Y):

$$\sigma_Y = \sqrt{\frac{\sum Y^2}{N} - \bar{Y}^2}$$

$$\sigma_Y = \sqrt{\frac{184}{7} - (4.286)^2} \approx \sqrt{26.2857 - 18.3697} \approx \sqrt{7.916} \approx 2.814$$

Coefficient of Variation (C.V._Y):

$$C.V._Y = \frac{2.814}{4.286} \times 100 \approx \mathbf{65.66}$$

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3. Conclusion on Consistency

Item	Mean ($\bar{X}$)	Standard Deviation (σ)	C.V.
Guavas	4.286	1.030	24.04%
Oranges	4.286	2.814	65.66%

Since the **Coefficient of Variation for Guavas (24.04%)** is **significantly lower** than the Coefficient of Variation for Oranges (65.66%), the consumption of **Guavas is more consistent** than the consumption of Oranges.

The next question in your list is **Question 28**.

28. Prove the Frequency Distribution Curve is Leptokurtic

To prove that the frequency distribution is **leptokurtic**, we must calculate the **Coefficient of Kurtosis (β_2)** and show that it is greater than 3 ($\beta_2 > 3$).

Class	Frequency (f)
10-15	1
15-20	4
20-25	8
25-30	19
30-35	35
35-40	20

Class	Frequency (f)
40-45	7
45-50	5
50-55	1

Step 1: Calculate Moments About an Assumed Mean (A)

We use the **step deviation method** to find the second (μ_2) and fourth (μ_4) central moments.

Let the Assumed Mean A be the midpoint of the class 30-35: $A = 32.5$.

The class width is $h = 5$.

Let $d' = \frac{X-32.5}{5}$.

Class	f	Midpoint (X)	d'	fd'	fd' ²	fd' ³	fd' ⁴
10-15	1	12.5	-4	-4	16	-64	256
15-20	4	17.5	-3	-12	36	-108	324
20-25	8	22.5	-2	-16	32	-64	128
25-30	19	27.5	-1	-19	19	-19	19
30-35	35	32.5 (A)	0	0	0	0	0
35-40	20	37.5	1	20	20	20	20
40-45	7	42.5	2	14	28	56	112
45-50	5	47.5	3	15	45	135	405
50-55	1	52.5	4	4	16	64	256
Total	= 100			$\sum fd' = 2$	$\sum fd'^2 = 212$	$\sum fd'^3 = 22$	$\sum fd'^4 =$

Moments about A (in units of d'):

- $\mu'_1 = c = \frac{\sum fd'}{N} = \frac{2}{100} = 0.02$

- $\mu'_2 = \frac{\sum fd'^2}{N} = \frac{212}{100} = 2.12$
- $\mu'_4 = \frac{\sum fd'^4}{N} = \frac{1520}{100} = 15.20$

σ

Step 2: Convert to Central Moments (μ_2 and μ_4)

1. Second Central Moment (μ_2)

$$\begin{aligned}\mu_2 &= h^2[\mu'_2 - c^2] \\ \mu_2 &= 5^2[2.12 - (0.02)^2] = 25[2.12 - 0.0004] \\ \mu_2 &= 25 \times 2.1196 = \mathbf{52.99}\end{aligned}$$

2. Fourth Central Moment (μ_4)

$$\mu_4 = h^4[\mu'_4 - 4\mu'_3c + 6\mu'_2c^2 - 3c^4]$$

We need μ'_3 : $\mu'_3 = \frac{\sum fd'^3}{N} = \frac{22}{100} = 0.22$

$$\begin{aligned}\mu_4 &= 5^4[15.20 - 4(0.22)(0.02) + 6(2.12)(0.02)^2 - 3(0.02)^4] \\ \mu_4 &= 625[15.20 - 0.0176 + 0.005088 - 0.00000048] \\ \mu_4 &= 625 \times 15.18748752 \approx \mathbf{9492.1797}\end{aligned}$$

σ

Step 3: Calculate the Coefficient of Kurtosis (β_2)

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

$$\beta_2 = \frac{9492.1797}{(52.99)^2} = \frac{9492.1797}{2807.9401} \approx \mathbf{3.379}$$

Conclusion

Since the **Coefficient of Kurtosis** ($\beta_2 \approx 3.379$) is **greater than 3**, the frequency distribution curve is indeed **leptokurtic**.

The next question in your list is **Question 29**. This question has two parts, (a) and (b).

29. Test of Skewness and Kurtosis (Two Parts)

Part (a): Central Moments Given

The first four **central moments** (moments about the mean, μ_k) are given:

$$\mu_1 = 0, \quad \mu_2 = 2.3, \quad \mu_3 = 0.9, \quad \mu_4 = 15.65$$

1. Test of Skewness

We use the moment coefficient of skewness, β_1 :

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

$$\beta_1 = \frac{(0.9)^2}{(2.3)^3} = \frac{0.81}{12.167} \approx \mathbf{0.0666}$$

Since $\beta_1 > 0$, the distribution is **positively skewed**.

2. Test of Kurtosis

We use the moment coefficient of kurtosis, β_2 :

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

$$\beta_2 = \frac{15.65}{(2.3)^2} = \frac{15.65}{5.29} \approx \mathbf{2.958}$$

3. Discuss the Nature of the Curve

- **Skewness:** Since β_1 is a small positive value (0.0666), the distribution is slightly **Positively Skewed** (long right tail).
- **Kurtosis:** Since β_2 is less than 3 ($2.958 < 3$), the distribution is **Platykurtic** (flatter peak and thinner tails than a normal distribution).

σ

Part (b): Moments About an Arbitrary Point ($A = 4$)

The first four moments about $x = 4$ (μ'_k) are:

$$\mu'_1 = c = 1, \quad \mu'_2 = 4, \quad \mu'_3 = 10, \quad \mu'_4 = 45$$

1. Calculate Moments About the Mean (μ_k)

We use $c = \mu'_1 = 1$.

- $\mu_1 = 0$
- $\mu_2 = \mu'_2 - c^2 = 4 - (1)^2 = \mathbf{3}$
- $\mu_3 = \mu'_3 - 3\mu'_2c + 2c^3 = 10 - 3(4)(1) + 2(1)^3 = 10 - 12 + 2 = \mathbf{0}$
- $\mu_4 = \mu'_4 - 4\mu'_3c + 6\mu'_2c^2 - 3c^4 = 45 - 4(10)(1) + 6(4)(1)^2 - 3(1)^4 = 45 - 40 + 24 - 3 = \mathbf{26}$

2. Calculate Characteristics

- **Mean ($\bar{X}$):** $\bar{X} = A + c = 4 + 1 = \mathbf{5}$
- **Variance (σ^2):** $\mu_2 = \mathbf{3}$
- **Standard Deviation (σ):** $\sqrt{\mu_2} = \sqrt{3} \approx \mathbf{1.732}$

3. Comment upon the Nature of the Distribution

- **Skewness (β_1):**

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{0^2}{3^3} = 0$$

Since $\beta_1 = 0$, the distribution is **Symmetrical**.

- **Kurtosis (β_2):**

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{26}{3^2} = \frac{26}{9} \approx 2.889$$

Since $\beta_2 < 3$, the distribution is **Platykurtic**.

Characteristic	Value	Comment on Nature
Mean ($\bar{X}$)	5	
Variance (σ^2)	3	
μ_3	0	
β_1 (Skewness)	0	Symmetrical
β_2 (Kurtosis)	2.889	Platykurtic

The next and final question in your list is **Question 30**.

30. Calculate Bowley's Coefficient of Skewness

To calculate **Bowley's Coefficient of Skewness** (S_B), we need to find the lower quartile (Q_1), the median (Q_2), and the upper quartile (Q_3).

The formula is:

$$S_B = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

Step 1: Prepare the Cumulative Frequency Table

Distance (km)	Frequency (f)	Cumulative Frequency (c.f.)
0-10	2	2
10-20	5	7
20-30	10	17
30-40	15	32
40-50	10	42
50-60	4	46
60-70	1	47
Total	= 47	

Step 2

Step 2: Calculate Quartiles (Q_1, Q_2, Q_3)

The general formula for quartiles is:

$$Q_k = L + \frac{\frac{kN}{4} - \text{c.f.}_{\text{precedin}}}{f_{Q_k}} \times h$$

The total frequency is $N = 47$, and the class width $h = 10$.

A. Lower Quartile (Q_1)

- **Position:** $N/4 = 47/4 = 11.75$
- **Q_1 Class:** 20 – 30 (c.f. is 17)
- **Parameters:** $L = 20$, $\text{c.f.}_{\text{precedin}} = 7$, $f_{Q_1} = 10$, $h = 10$.

$$Q_1 = 20 + \frac{11.75 - 7}{10} \times 10 = 20 + 4.75 = 24.75$$

B. Median (Q_2)

- **Position:** $2N/4 = N/2 = 47/2 = 23.5$
- **Q_2 Class:** 30 – 40 (c.f. is 32)
- **Parameters:** $L = 30$, c.f._{precedin} = 17, $f_{Q_2} = 15$, $h = 10$.

$$Q_2 = 30 + \frac{23.5 - 17}{15} \times 10 = 30 + \frac{6.5}{15} \times 10$$

$$Q_2 = 30 + 4.3333 \approx \mathbf{34.33}$$

C. Upper Quartile (Q_3)

- **Position:** $3N/4 = 3(47)/4 = 35.25$
- **Q_3 Class:** 30 – 40 (c.f. is 32)
- **Parameters:** $L = 30$, c.f._{precedin} = 17, $f_{Q_3} = 15$, $h = 10$.

$$Q_3 = 30 + \frac{35.25 - 17}{15} \times 10 = 30 + \frac{18.25}{15} \times 10$$

$$Q_3 = 30 + 12.1667 \approx \mathbf{42.17}$$

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Step 3: Calculate Bowley's Coefficient of Skewness (S_B)

Substitute the quartile values into the formula: $Q_1 = 24.75$, $Q_2 = 34.33$, $Q_3 = 42.17$.

$$S_B = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

$$S_B = \frac{42.17 + 24.75 - 2(34.33)}{42.17 - 24.75}$$

$$S_B = \frac{66.92 - 68.66}{17.42}$$

$$S_B = \frac{-1.74}{17.42} \approx \mathbf{-0.0999}$$

Bowley's Coefficient of Skewness is approximately -0.10. Since the value is negative, the distribution is slightly **negatively skewed**.